

# Technology Stack

|               |                                                      |
|---------------|------------------------------------------------------|
| Date          | 12 July 2026                                         |
| Team ID       | 6a2fbf036b1ed0bd8fa789aa                             |
| Project Name  | AI Powered Debt Relief & Financial Recovery Platform |
| Maximum Marks | 2 Marks                                              |

## Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

## Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen       | Justification / Purpose                                                                                                                                                                                      |
|------|--------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | React.js + Vite         | Component-based UI enables reusable dashboard widgets (loan cards, charts); Vite provides fast builds and hot-reload for rapid iteration on a responsive, borrower-facing interface.                         |
| 2    | Backend / Server-Side          | Python + FastAPI        | FastAPI offers high-performance async request handling, automatic OpenAPI docs, and built-in data validation — well suited for exposing authentication, financial calculation, and AI negotiation endpoints. |
| 3    | Database / Data Storage        | SQLite + SQLAlchemy ORM | Lightweight, zero-configuration database ideal for development and moderate scale; SQLAlchemy ORM abstracts queries and allows easy migration to PostgreSQL/MySQL as the platform grows.                     |
| 4    | Cloud / Hosting / Deployment   | Docker + Render/AWS     | Containerizing the FastAPI backend and React frontend with Docker ensures consistent environments across development and production, with easy, cost-effective scaling on cloud hosting.                     |
| 5    | Version Control & CI/CD        | GitHub + GitHub Actions | Enables seamless team collaboration through branching and pull requests, with automated testing and deployment pipelines triggered on every commit.                                                          |

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|------|--------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Third-Party APIs / Other Tools | Google Gemini API | Powers AI-driven negotiation letter generation and negotiation strategy suggestions based on the borrower's financial profile, with a rule-based fallback if the API is unavailable. |